

Fábio F. O. Silva

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Experience

- **QuintoAndar** **São Paulo, Brazil (Remote)**
Senior Software Engineer, Full Time *Feb 2025 – Present*
Designing and implementing AI-powered multi-agentic solutions to enhance and scale inside sales operations. Driving efficiency and customer engagement using Large Language Models (LLMs).
Tech Stack: AWS, Python, LLMs, AI Agents.
- **VTEX** **João Pessoa, Brazil**
Senior Software Engineer, Full Time *Oct 2020 – Jan 2025*
Part of the Payments team, responsible for maintaining a high-throughput, PCI-compliant Payment Gateway and related financial products. Executed stack modernization and infrastructure efficiency initiatives, reducing costs while improving system reliability and resiliency.
Tech Stack: AWS, .NET, C#, SQL, Redis, ActiveMQ.
- **Scontain U.G.** **Dresden, Germany**
Software Engineer, Full Time *Apr 2020 – Sep 2020*
Contributed to the core SCONE confidential computing runtime and co-developed a modified Intel SGX Linux kernel driver to support selective enclave eviction. Led the design and implementation of a Kubernetes device plugin/driver to enable SGX usage in containerized pods.
Tech Stack: C, Docker, Bash, Intel SGX, Python, Gitlab, CI/CD, Linux.
- **Distributed Systems Laboratory** **Campina Grande, Brazil**
Research Assistant & Software Developer *Jan 2017 – Mar 2020*
Developed a secure cloud execution ecosystem leveraging Intel SGX hardware-enforced security extensions. Built mechanisms to protect workloads against attacks from privileged system users (cloud operators/sysadmins) and compromised software layers (hypervisors).
Tech Stack: Docker, SCONE, OpenStack, Intel SGX, C++, MongoDB, Python, Apache Kafka, Git.
- **Technische Universität Dresden** **Dresden, Germany**
Research Assistant & Software Developer, Intern *Dec 2017 – Jan 2018*
Improved Python runtime support and performance within SCONE confidential enclaves. Modified CPython source code (C) and integrated the Intel Math Kernel Library (MKL) to run securely inside SGX environments.
Tech Stack: Docker, SCONE, Intel SGX, C, Python, musl libc, PySpeed, PyPy, Git.
- **PrivIoT** **Campina Grande, Brazil**
Software Developer *Jan 2016 – Dec 2016*
Researched privacy-by-design patterns for Internet of Things (IoT) architectures. Developed and validated reference IoT applications as case studies to verify secure software development guidelines.
Tech Stack: MongoDB, Python, Flask, Ionic, JavaScript, AngularJS, D3.js, Git.

- Analytics Lab**

Campina Grande, Brazil

- Data Analyst & Software Developer*

Aug 2014 – Dec 2015

Analyzed 3 years of agricultural cohort data from 200+ farmers, creating interactive data visualizations with JavaScript. Developed a GPS-enabled mobile application for the public transportation sector to map bus stops, validate routes, and enrich municipal datasets.

Tech Stack: R, D3.js, Chart.js, Android SDK, Git.
- Software Practices Laboratory**

Campina Grande, Brazil

- Software Developer & Research Assistant*

Nov 2012 – Jul 2014

Developed an Eclipse plug-in that automates the migration and optimization of Java Collections Framework (JCF) classes in source code. (Demo: <https://www.youtube.com/watch?v=PUIEP1VmI-Q>)

Tech Stack: Java, Eclipse, Eclipse JDT, Design Patterns, Git.
- 9Ideia Digital**

Campina Grande, Brazil

- Web & Android Developer, Intern*

Nov 2011 – Feb 2012

Developed web applications and custom Content Management Systems (CMS) using the LAMP stack (Linux, Apache, MySQL, PHP). Engineered Android applications utilizing the Android SDK as part of a technical internship.

Tech Stack: Apache, HTML, CSS, JavaScript, Java, Android SDK, jQuery, PHP, MySQL, Git.

Voluntary Experience

- CyanogenMod Project**

- Device Tree Maintainer*

Nov 2015 – Oct 2016

Added initial support to Sony Xperia Z3 Dual (D6633) on CyanogenMod project. It is also the device tree maintainer's role to perform code reviews of contributions made through commits on Gerrit Code Review system.

Tech Stack: Linux Kernel, Android Build Engine, AOSP, Android Architecture, C++, Makefile, Gerrit, Git.

Education

- Udacity, Inc.**

Remote

- C++ Nanodegree*

2021
- Federal University of Campina Grande**

Campina Grande, Brazil

- M.Sc. Computer Science, Information Security*

2018 – 2020
- Federal University of Campina Grande**

Campina Grande, Brazil

- B.Sc. Computer Science*

2012 – 2017
- Infogenius Technical College**

Campina Grande, Brazil

- Certificate in Information Technology*

2010 – 2012

Notable Projects

- **Kafka Proxy SGX**

- *Lead Developer, Software*

- A transparent, drop-in security proxy that encrypts in-transit and at-rest Kafka cluster data. Leverages confidential computing (Intel SGX) to protect sensitive data and encryption keys from infrastructure operators and host-level threats.

- **RunMusic App**

- *Founder & Developer, Android Application*

- An Android application that synchronizes music playback tempo with the runner's real-time pace. Awarded 1st place in the Entrepreneurship category of the III Campus Mobile national contest.

- **Java Compiler**

- *Academic Project, Compiler Course*

- Developed a Java-like compiler using JFlex (lexical analyzer) and CUP (LALR parser). The front-end parses standard Java syntax features, while the back-end compiles to target Assembly code.

Key Skills

- **Programming Languages:** Python, C# (.NET), C/C++, Java, JavaScript, SQL, R.

- **Cloud & Infrastructure:** AWS, Docker, Kubernetes, OpenStack, Linux, Git, CI/CD.

- **Databases & Middleware:** SQL, MongoDB, Redis, ActiveMQ, Apache Kafka.

Languages

- Portuguese (*native*)

- English (*professional proficiency*)

Awards

- **2015:** Best application in Entrepreneurship Category of III Campus Mobile national contest, promoted by Instituto Embratel-Claro.

- **2011:** 1st place on programming contest promoted by Infogenius Technical College.

Publications

S. Arnautov, A. Brito, P. Felber, C. Fetzer, F. Gregor, R. Krahn, W. Ozga, A. Martin, V. Schiavoni, **F. Silva**, M. Tenorio, and N. Thümmel. PubSub-SGX: Exploiting Trusted Execution Environments for Privacy-Preserving Publish/Subscribe Systems. In *2018 IEEE 37th Symposium on Reliable Distributed Systems (SRDS)*, pages 123–132, 10 2018.

I. Blanquer, F. Brasileiro, A. Brito, A. Calatrava, A. Carvalho, C. Fetzer, F. Figueiredo, R. Guimaraes, L. Marinho, W. Meira Jr, A. Silva, A. Alberich-Bayarri, E. Camacho-Ramos, A. Jimenez-Pastor, A. L. Ribeiro, B. Nascimento, and **F. Silva**. Federated and secure cloud services for building medical image classifiers on an intercontinental infrastructure. *Future Generation Computer Systems*, 110, 04 2020.

M. S. L. da Silva, **F. Silva**, and A. Brito. Squad: A Secure, Simple Storage Service for SGX-Based Microservices. In *2019 9th Latin-American Symposium on Dependable Computing (LADC)*, pages 1–9, 11 2019.

L. Sampaio, **F. Silva**, A. Souza, A. Brito, and P. Felber. Secure and Privacy-Aware Data Dissemination for Cloud-Based Applications. In *Proceedings of the 10th International Conference on Utility and Cloud Computing*, UCC '17, pages 47–56, New York, NY, USA, 2017. ACM.

F. Silva, M. Silva, and A. Brito. Kafkaproxy: data-at-rest encryption and confidentiality support for kafka clusters. In *SBSeg'2020, XX Simpósio Brasileiro em Segurança da Informação e de Sistemas Computacionais*, 10 2020.